KEITHLEY 2450

Studio

User Manual

Professional I-V Characterisation · FET Analysis · Q-Point Computation
Digital Multimeter · Cycling & Stability · Bias Stress · Parameter Extraction

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1. Introduction

Keithley 2450 Studio is a professional desktop application for instrument-controlled electrical characterisation using the Keithley 2450 Source Measure Unit (SMU). It provides a complete workflow from instrument connection through measurement execution to data export across three dedicated measurement modes.

1.1 Key Capabilities

- Voltage and current sweeps — linear, logarithmic, and arbitrary list-based
- Dual sweep (forward + reverse) for hysteresis analysis
- Pulse mode with configurable on/off timing per point
- Stepper mode — Node 2 steps a bias while the master SMU sweeps, producing a family of curves
- Cycling & Stability — repeat sweeps N times with configurable rest, gradient colouring, Mean±σ band overlay
- Bias stress — apply a fixed voltage between cycles while logging current drift over time
- Ghost overlay — previous full run displayed as faded grey reference behind new live data
- Compare CSV Manager — load historical CSVs as dashed reference overlays; toggle, rename, recolour
- Parameter extraction — Vth, SS (mV/dec), Ion/Ioff, hysteresis area; cycle trend (ΔVth, Ion drift)
- Measurement quality — per-point averaging (1–16×), settling check, compliance pre-check before sweep
- FET Transfer Curve (Id vs Vgs) and Output Curve (Id vs Vds) via TSP-Link dual-SMU
- Q-Point analysis — gm at gate bias (Transfer); load line + Pd hyperbola (Output)
- Digital Multimeter — DC Voltage, DC Current, 2W/4W Resistance, Diode, Continuity, Power
- Auto-save CSV and PNG on sweep completion
- Dark and light theme with persistent preference

1.2 System Requirements

Component	Requirement
Operating System	Windows 10 / 11 (64-bit)
Python	3.9 or later
VISA Runtime	NI-VISA 21+ or Keysight IO Libraries Suite
Instrument	Keithley 2450 SMU (firmware 1.7.0+)
FET / Stepper Mode	Second Keithley 2450 + TSP-Link RJ-45 cable
Display	1280 × 720 minimum, 1920 × 1080 recommended
Python Packages	pyvisa, pyvisa-py, numpy, matplotlib

1.3 Installation

1. Install the NI-VISA runtime from ni.com/en/support/downloads/drivers/download.ni-visa.html
2. Install Python 3.9 or later from python.org — tick Add Python to PATH during installation
3. Install Python dependencies: pip install pyvisa pyvisa-py numpy matplotlib
4. Run the application: python keithley_pro_final.py

5. To build a standalone .exe: pyinstaller --onefile --windowed --name Keithley_LabView keithley_pro_final.py

> ℹ Note: The VISA address must match exactly what NI-MAX or Keysight Connection Expert shows, e.g. USB0::0x05E6::0x2450::04465297::INSTR

2. Mode Selector & Application Launch

On launch the application displays a splash screen while libraries are loaded, then opens a full-screen Mode Selector. Three measurement modes are available as clickable tiles.

Mode	Icon	Description
IV — I-V Characterisation	⚡	Voltage/current sweeps, cycling, curve comparison, parameter extraction
FET — FET Characterisation	🔬	Transfer & output curves, bias stress, Q-point analysis
DMM — Digital Multimeter	📊	Live voltage/current readout for probe landing and contact checking

Click the Open button on any tile to launch that mode. Closing a mode window returns to the Mode Selector. Pressing Escape on the selector or clicking ✕ Quit exits the application.

Settings (VISA address, save folder, sample ID, dark/light theme) are remembered in keithley_config.json next to the script between sessions.

3. I-V Characterisation Window (IV Mode)

The IV window opens maximised. It is divided into the top header bar, the left scrollable control panel, the right chart area with axis selector sidebar, and a status bar at the bottom. Changes to any control take effect on the next RUN.

3.1 Header Bar

Element	Description
KEITHLEY 2450 title	Application identifier — left side
Sample ID	Short identifier for the DUT — used in auto-saved filenames and CSV headers
Save Folder	Directory for auto-saved CSV and PNG files — click ... to browse
VISA Address	Full VISA resource string for the master SMU — editable at any time
☀ / 🌙 toggle	Switch between light and dark theme — preference saved automatically
● SMU: ...	Live connection status — green = connected (shows serial number), red = disconnected. Polls every 3 s when idle.

3.2 Status Bar

State	Colour	Meaning
Ready	Green	Instrument idle, ready to sweep
Running	Amber	Sweep in progress
Stopping	Red	Stop requested, waiting for instrument abort
Complete	Green	Sweep finished, data auto-saved
Error	Red	Instrument or communication error
Stopped	Red	Sweep halted by user

3.3 Source & Sweep Card

3.3.1 Source Mode

Mode	Typical Use
Voltage Sweep	Diode, resistor, semiconductor I-V
Current Sweep	LED characterisation
Voltage List Sweep	Non-uniform voltage steps from a custom list
Current List Sweep	Non-uniform current steps from a custom list
Voltage Bias	Single fixed voltage — leakage, standby current
Current Bias	Single fixed current — Vf at fixed drive

3.3.2 Option Checkboxes

- Stepper (Node 2): Node 2 steps through a range of bias voltages; the full sweep runs at each bias step, producing a family of curves. Requires TSP-Link.

- Dual Sweep: Runs sweep in both directions (e.g., 0 → 5 V then 5 → 0 V) to capture hysteresis. Both passes are stored as a single curve.
- Pulse Mode: Replaces DC sweep with a pulsed sweep. Requires On Time and Off Time parameters.

3.3.3 Sweep Parameters

Start, Stop, Step, and Points are linked. Editing any three auto-updates the fourth: Points = |Stop − Start| / |Step| + 1. Editing Points back-calculates Step.

Parameter	Description
Start	First source value (V or A)
Stop	Last source value (V or A)
Step	Increment between points
Points	Total number of measurement points — auto-calculated from Step
Sweep Type	Linear: uniform spacing. Logarithmic: log-spaced points — useful for impedance sweeps spanning decades.

3.3.4 List Sweep Values

When a List Sweep mode is selected, the Start/Stop/Step area is replaced by an inline table. Double-click any row to edit inline. Use + Row, ✕ Delete, and ⬆ Import (CSV or Excel) to manage values.

3.3.5 Pulse Parameters

Visible only when Pulse Mode is checked.

Parameter	Description
Bias Level	Voltage applied between pulses — the off-state level
On Time (s)	Duration of each pulse — time at the sweep level per point
Off Time (s)	Duration between pulses — recovery / off time

3.3.6 Source Range, Compliance, and Source Delay

Setting	Behaviour
Best Fixed	Smallest fixed range covering the full sweep — fastest settling (recommended)
Auto	Auto-ranges point by point — handles wide dynamic range but slower
Fixed range (200mV / 2V / 20V / 200V)	Forces a specific range — use when Best Fixed selects an oversized range

⚠ Warning: If measured current is flat at an unexpected level the device has hit compliance. Raise the Compliance value or reduce the sweep range.

3.3.7 Stepper — Node 2 Configuration

Visible only when Stepper is checked. Requires a second 2450 connected via TSP-Link (Master = Node 1, Slave = Node 2).

Parameter	Description

Start (V)	First bias voltage applied by the Slave SMU (Node 2)
Stop (V)	Last bias voltage applied by the Slave SMU
Points	Number of equidistant bias steps between Start and Stop

3.4 Measure Card

3.4.1 Current

- Range: Auto uses auto-ranging; a fixed range gives faster and more stable measurements at that scale.
- Min Auto Range: Prevents the instrument from auto-ranging below this value — avoids slow transitions on near-zero readings.

3.4.2 Voltage

- Programmed: reports the commanded voltage. Actual: reports the ADC-measured terminal voltage — differs for 4-Wire sense.

3.4.3 Resistance & Power

- Resistance: computed as |V/I| per point. Enable for R-V curves.
- Power (W): computed as |V×I| per point.
- Include Timestamp: adds elapsed time (seconds since sweep start) as the first CSV column.

3.5 Instrument Settings Card

Parameter	Description
NPLC	Power Line Cycles per measurement. Higher = more averaging, slower, less noise. 1 NPLC = 20 ms at 50 Hz.
Auto Zero	Once: zeros ADC before the sweep (recommended). Off: fastest. On: zeros before every point.
Input Jacks	Rear or Front — must match physical wiring.
Sense Mode	2-Wire: standard. 4-Wire (remote sense): eliminates lead resistance — required for sub-10 Ω devices.
Output OFF State	Normal / High-Z / Zero — behaviour when sweep ends or is stopped. High-Z disconnects output completely.
High Capacitance	Enable for devices with capacitance > 10 nF to avoid false compliance tripping.

3.6 Cycling & Stability Card

The Cycling & Stability card allows the sweep to be repeated N times automatically, with optional rest periods and user-controlled triggering between cycles. This is the primary tool for stability testing and bias-stress characterisation.

3.6.1 Basic Cycling Parameters

Parameter	Description
Enable Cycling	Activates N-cycle repeat mode
Cycles	Number of complete sweeps to perform

Rest (s)	Pause duration between cycles — instrument output remains on during rest
Trigger	Auto: proceeds automatically after rest. Manual: shows ▶ Next Cycle button and waits for user press before each subsequent cycle.

3.6.2 Cycle Names

The Cycle Names text box accepts one label per line. If filled, names are used automatically in the legend, CSV headers, and status labels (e.g. Fresh, +20°C, Stressed, Recovery). If left empty, cycles are labelled Cyc 1, Cyc 2, etc.

3.6.3 Display Options

Option	Behaviour
Gradient	Each cycle is plotted in a distinct colour from the palette for easy visual separation
Show all	All completed cycles remain visible simultaneously on the chart
Mean±σ	After 2 or more cycles complete, a blue Mean line and shaded ±1σ band are overlaid across all cycles

3.6.4 Ghost Overlay

The Ghost: ON / OFF button controls whether the previous full run is preserved as a faded grey reference behind the live data. When a new RUN is started, the existing data is deep-copied as the ghost before being cleared. The ghost persists across cycles within the same session.

✓ Tip: Use Ghost Overlay to visually compare before/after stress or between sample batches without loading a CSV.

3.7 Measurement Quality Card

These settings improve measurement accuracy at the cost of speed. They are applied before and during each sweep.

Setting	Description
Avg / point (1–16×)	Each data point is the mean of N consecutive readings. NPLC is scaled proportionally to match the equivalent integration time.
Settling check	After averaging, the instrument re-reads until two consecutive averaged values agree within the Tolerance %. Prevents capturing transient values on capacitive devices.
Tolerance (%)	Convergence criterion for the settling check — typical value 1.0%
Compliance pre-check	Before starting the sweep, the instrument measures current at the Start voltage. If it is within 5% of compliance the sweep is aborted with a warning — prevents damage from shorted devices.

3.8 Chart Area, Toolbar & Axis Selector

3.8.1 Toolbar Buttons

Button	Action
Clear Plot	Clears all sweep and cycle data — prompts for confirmation

Export CSV	Saves current dataset to CSV (long format for single run; wide format for cycling)
Cycle CSV	Saves all cycles in wide format — one column block per cycle
Save PNG	Saves chart as 300 DPI PNG — always in light mode regardless of current theme
Auto Scale	Resets both axes to fit all visible data
Ghost: ON/OFF	Toggles the ghost overlay display
⚗ Analysis	Runs parameter extraction on the most recent data
📊 Compare CSV	Opens the Compare CSV Manager panel

3.8.2 Progress Bar

The progress bar fills 0–100% as data is acquired. The percentage label and elapsed time (MM:SS) update in real time. During cycling, progress reflects the total across all cycles.

3.8.3 Live Chart

New data points appear on the chart as they are acquired — the chart refreshes every ~5% of sweep progress. Axes use SI engineering notation (e.g., 10 mA instead of 0.01 A). During cycling, each cycle is plotted in a distinct colour. The legend identifies each curve by channel name, cycle name, and stepper bias if active.

3.8.4 Axis Selector Sidebar

Section	Control	Description
X-Axis	Radio buttons	Select one channel as the horizontal axis. Default: Voltage (V).
Y1-Axis (left)	Checkboxes	One or more channels on the left vertical axis. Default: Current (A).
Y2-Axis (right)	Checkboxes	Secondary right vertical axis — useful for Resistance or Power alongside Current.

Available channels for all axes: Voltage (V), Current (A), Time (s), Resistance (Ω), Power (W).

3.9 Parameter Extraction Panel

The Parameter Extraction panel is collapsed by default — click the header bar to expand it. It operates on the most recently completed sweep or the last completed cycle.

Extractor	Method
Vth	Linear extrapolation of √Id vs V at the point of maximum transconductance gm
SS (mV/dec)	Subthreshold swing — inverse of the maximum slope of log10(Id) vs V in the subthreshold region
Ion/Ioff	Ratio of maximum to minimum |Current| in the sweep range
Hysteresis	Signed area between forward and reverse sweep halves (trapezoid integration) — requires Dual Sweep

Results appear inline next to the Run Analysis button. When cycling is active and more than one cycle has completed, trend values are also shown:

- ΔVth — change in threshold voltage from first to last cycle
- Ion drift — percentage change in peak current from first to last cycle

ℹ Note: Parameter extraction on an IV sweep is most meaningful for thin-film transistors (TFTs) and organic semiconductors measured in Transfer-style sweeps (gate as source, drain on Y-axis). For standard diode/resistor sweeps the extractors will return N/A.

3.10 Compare CSV Manager

The Compare CSV Manager loads historical CSV files and overlays them as dashed reference lines on the live chart. This allows direct visual comparison between runs without importing data into a separate tool.

Action	Description
+ Add CSV(s)	Open file dialog to select one or more CSV files
👁 toggle	Show or hide individual reference curves
Label field	Click to rename — shown in the legend
● colour swatch	Click to cycle through the 10-colour reference palette
✕ remove	Remove a reference from the overlay
✕ Clear All	Remove all references

The Compare Manager parses both the long format (single-run CSV) and the wide format (cycling CSV) produced by this application. It also accepts any CSV with a Voltage column and one or more Current / (A) columns.

4. FET Characterisation Window (FET Mode)

The FET Characterisation window provides a dedicated two-SMU measurement environment for MOSFET characterisation. Gate, Drain, and Source terminals each have independent force and measure configuration. The window opens maximised with the parameter table on the left and the chart on the right.

⚠ Warning: Both SMUs must be connected via TSP-Link RJ-45 cable (rear panel TSP-LINK port). Set Node IDs before use: Master = 1, Slave = 2 via MENU → System → TSP-Link Node on the front panel of each unit.

4.1 Measurement Mode

Mode	Description
Transfer Curve (Id vs Vg)	Gate sweeps; Drain held at bias. X-axis = Vg. Each curve labelled Vds = X V.
Output Curve (Id vs Vds)	Drain sweeps; Gate held at bias steps. X-axis = Vds. Each curve labelled Vg = X V.

Switching mode automatically reconfigures the Gate and Drain Operation Mode dropdowns to sensible defaults.

4.2 Parameter Table

The table has one column per terminal (Gate, Drain, Source) and one row per parameter. Column headers are colour-coded: Gate = green, Drain = blue, Source = pink. Parameters are greyed out when a terminal is assigned to NONE or GNDU.

4.2.1 Instrument Assignment

Assign each terminal to one of: NONE, GNDU, SMU1 (Node 1 — Master), SMU2 (Node 2 — Slave). Assigning NONE or GNDU disables all force/measure fields for that column.

Default Assignment	Terminal
SMU1 (Node 1 — Master)	Gate
SMU2 (Node 2 — Slave)	Drain
GNDU	Source

4.2.2 Operation Mode

Operation Mode	Description
Voltage Bias	Terminal held at a fixed voltage (set in the Bias row)
Voltage Linear Sweep	Terminal sweeps Start → Stop with Step spacing
Voltage List Sweep	Terminal steps through an arbitrary list of voltages (click Edit… to manage)
Voltage Log Sweep	Terminal sweeps logarithmically between Start and Stop
Voltage Step	Linear sweep — used for the bias terminal in Output mode to produce a family of curves
Current Bias / Sweep variants	Same as voltage variants but sources current
Common	Terminal tied to chassis ground through the SMU

4.2.3 Points — Auto and Manual

Each terminal has an Auto/Manual toggle in the Points row:

- Auto (ticked, default): Points = |Stop − Start| / |Step| + 1, updated live. Entry field is greyed; blue auto badge is shown. Start/Stop/Step changes auto-recalculate.
- Manual (unticked): Entry field is editable. Sweep uses linspace(Start, Stop, N). Step is used only to display auto-points and is ignored for the actual sweep.

4.2.4 Dual Sweep

Each terminal has a Dual Sweep checkbox. When ticked, the sweep runs in both directions (Start → Stop → Start reversed) for that terminal. Useful for hysteresis characterisation on Transfer Curves.

4.2.5 Compliance, Source Delay, Power-On Delay, and OVP

Terminal	Recommended Compliance	Rationale
Gate	10 mA (0.01 A)	Gate oxide protection — prevents breakdown during transient ringing
Drain	100 mA – 1 A	Set slightly above the maximum expected Id for the gate range used
Source	100 mA	Source is GNDU; this rarely matters unless measuring Source current

✓ Tip: Source Delay for power MOSFETs (e.g. IRFP240: Ciss ≈ 1400 pF): use 0.10–0.15 s to allow full gate-charge settling. For small-signal MOSFETs 20–50 ms is sufficient.

Power-On Delay: additional settling time applied once at the start of the sweep before any measurement is taken. Useful for devices that need time to stabilise after bias is applied.

Overvoltage Protection (OVP): hardware clamp at 20 V, 40 V, 60 V, 100 V, or 200 V. OFF by default. Enable when using long probe cables near high-voltage equipment.

4.2.6 Measure Settings

Setting	Description
Current Measure	Check to record drain current. Column Name is used in CSV headers.
Current Range	Limited Auto: auto-ranges but stays above a floor value. Fixed ranges for fast, stable measurements.
Voltage Measure	Check to record terminal voltage. Programmed = commanded; Actual = ADC-measured.
Column Name	Custom header string for the CSV — default GateV / DrainV / SourceV
NPLC	Integration time per point for that terminal. Apply to all button propagates one value to all three terminals.

4.3 Cycling & Bias Stress Panel

The Cycling & Bias Stress panel is collapsible (click header to expand). It provides the same cycling capability as the IV window plus a dedicated bias stress phase between cycles.

4.3.1 Cycling

Parameter	Description
Enable Cycling	Activates N-cycle repeat mode
Cycles	Number of full sweeps to perform
Rest (s)	Idle pause between cycles — outputs remain on
Show all cycles	All completed cycles remain on chart simultaneously
Gradient colors	Each cycle plotted in a distinct colour
Mean±σ band	Blue mean line and ±1σ shaded band shown after 2 or more cycles
Ghost overlay	Previous full run preserved as faded grey reference

Cycle Names text box: one label per line. Auto-used if filled — otherwise Cyc 1, Cyc 2, etc.

4.3.2 Bias Stress

Bias stress is applied between cycles (not during the sweep). It holds a fixed voltage on the selected terminal for a configurable duration, logging drain current at regular intervals. This simulates accelerated device aging or threshold voltage shift under sustained gate/drain bias.

Parameter	Description
Enable Stress	Activates bias stress between cycles — requires cycling to be enabled and N > 1
Terminal	Gate or Drain — which terminal receives the stress voltage
Stress V	Fixed voltage applied during the stress phase
Duration (s)	Total stress duration in seconds
Measure every (s)	Interval at which drain current is sampled and displayed in the status bar during stress

⚠ Warning: Bias stress applies the stress voltage immediately after the previous sweep completes and before the rest period. Ensure compliance is set appropriately — the SMU will source the stress voltage until Duration expires or STOP is pressed.

4.4 Parameter Extraction Panel (FET)

The Parameter Extraction panel is collapsible. It operates on the most recently completed Transfer or Output measurement. For Output Curves, Vth and SS are not computed (they require a gate sweep).

Extractor	Method / Notes
Vth	Linear extrapolation of √Id vs Vgs at maximum transconductance. Transfer Curve only.
SS (mV/dec)	Subthreshold swing from the steepest portion of the log10(Id) vs Vgs curve. Transfer only.
Ion/Ioff	Ratio of maximum to minimum |Id| across the full sweep range.
Mobility (μFE)	Field-effect mobility: μFE = (L / W·Cox·Vds) × max(dId/dVgs). Requires Cox, W, L, and Vds_lin to be entered. Transfer only.
Hysteresis	Signed area between forward and reverse sweep halves. Requires Dual Sweep to be active.

Cycle trend summary (shown when 2 or more cycles completed):

- ΔVth — threshold voltage shift from first to last cycle (Transfer mode only)

- Ion drift — percentage change in peak drain current across cycles

4.5 Q-Point Analysis

The Q-Point panel sits between the toolbar and the chart. Tick Enable to activate — no new measurement is required. Q-point computation updates every time the chart is redrawn.

4.5.1 Transfer Curve Q-Point

Marks a specific gate voltage operating point and computes gm (transconductance) at that point.

Input	Description
Vgs_Q	Gate-to-source quiescent voltage at which to evaluate the Q-point (Volts)

Output on chart:

- Gold star marker (★) at (Vgs_Q, Id_Q)
- Dashed yellow crosshair lines from both axes to the marker
- Annotation box showing: Vgs, Id, and gm (Siemens)
- gm computed as dId/dVgs using numpy.gradient on the measured curve

4.5.2 Output Curve Q-Point

Draws a DC load line, finds its intersection with the selected Vg curve, and plots the constant-dissipation hyperbola through the operating point.

Input	Description
Vdd	Supply voltage — x-axis intercept of the load line (Id = 0 at Vds = Vdd)
RL	Load resistance (Ω) — sets the slope. Id_max = Vdd / RL at Vds = 0.
Vg curve	The gate bias curve to intersect — auto-populated from measured data

Output on chart:

- Dash-dot gold load line from (Vdd, 0) to (0, Vdd/RL) with intercept labels
- Gold star marker (★) at the intersection point (Vds_Q, Id_Q)
- Red dotted Pd hyperbola: Id = Pd_Q / Vds through the Q-point (equal-dissipation locus)
- Annotation box: Vg curve label, Vds_Q, Id_Q, Pd_Q (watts)

⚠ Warning: The Q-point is found at the last (saturation-region) intersection of the load line with the selected Vg curve. If no intersection is found (e.g. Vdd is smaller than the sweep range), 'No Q-point intersection' appears in the result bar.

5. Digital Multimeter (DMM Mode)

The DMM window provides live instrument readings for probe landing, contact verification, and quick spot measurements. It opens maximised. Click ► Connect to begin polling; the display updates every 400 ms.

5.1 Measurement Modes

Mode	Key	Description
DC Voltage	DCV	Sources 0 V, measures terminal voltage
DC Current	DCI	Sources 0 A, measures current — insert SMU in series
Resistance 2W	R2W	Sources test voltage, measures I, computes R = V/I — includes lead resistance
Resistance 4W	R4W	Kelvin sense — eliminates lead resistance. Required for R < 10 Ω.
Diode Test	DIODE	Sources constant current, measures forward voltage Vf
Continuity	CONT	Sources 0.5 V, displays CLOSED (< 10 Ω) or OPEN (≥ 10 Ω) in green/red
Power (VxI)	PWR	Sources voltage, measures current, displays P = V×I

5.2 Settings Bar

Setting	Description
VISA	Full VISA resource string for the SMU
Source V / I	Test source level — changes label to match the selected mode
Sense	2-Wire or 4-Wire — 4-Wire is forced automatically for R4W mode
Rate	Fast (0.1 PLC) / Medium (1 PLC) / Slow (10 PLC)
Terminals	Rear or Front — must match physical wiring
ℹ Wiring Guide	Opens a scrollable dialog with connection diagrams for every mode

5.3 Display

The main display shows the primary measurement value in large text. The secondary line shows related quantities (e.g., for DCV: current and resistance are also shown). Four tile readouts in the right column always show V, I, R, and P simultaneously.

5.4 Controls

Control	Action
HOLD	Freezes the main display — sub-label shows ▌ HOLD. New readings continue to update Min/Max.
MIN/MAX	Shows the minimum and maximum measured values since the last RESET
RESET	Clears Min/Max tracking
■ Disconnect	Turns off instrument output and closes the VISA session

✕ **Close**	Disconnects and closes the DMM window — returns to Mode Selector

ℹ Note: The DMM auto-disconnects after 3 consecutive read failures and shows '● SMU: Disconnected — not responding' in the status label. Reconnect using ► Connect.

6. Data Export

6.1 Auto-Save

On successful sweep completion, the application automatically saves a CSV and a PNG to the Save Folder. The filename is: <SampleID>_<YYYYMMDD>_<HHMMSS>.csv / .png. PNGs are always saved in light mode regardless of the active theme.

6.2 IV Window — Long-Format CSV (single run)

Used when cycling is disabled or only one cycle was run.

```
# Keithley 2450  Studio
# Sample ID,  DUT_001
# Date/Time,  05/03/2026  14:32:17
# Mode,  Voltage Sweep
# Extracted,  Vth = 2.14 V    SS = 87 mV/dec

Time (s), Voltage (V), Current (A), Resistance (Ω), Power (W)
0.000000, 0.000000000, 0.000012450, ...
```

6.3 IV Window — Wide-Format CSV (cycling enabled, N > 1)

When cycling is enabled and more than one cycle is completed, Export CSV and the auto-save use a wide format: one Voltage column followed by one current (and optionally R and P) column group per cycle. Cycle names appear as column headers.

```
# Cycles,  5

Voltage (V), Fresh  I (A), +20°C  I (A), Stressed  I (A), ...
0.0,  1.23e-12,  1.24e-12,  1.31e-12,  ...
```

The Cycle CSV button always exports in wide format regardless of the current cycle count.

6.4 FET Window CSV

```
# Keithley 2450  Studio  –  FET Characterisation
# Mode,  Transfer Curve (Id vs Vg)
# Gate,  SMU1 (Node 1 – Master),  Voltage Linear Sweep

Vg (V),  Id  Vds = 1.5 V (A),  Id  Vds = 3.5 V (A)
0.000000,  1.2e-11,  1.5e-11
```

FET Cycle CSV: same wide format as IV cycling — one X column, then one Id column per cycle per bias curve.

7. Typical Measurement Workflows

7.1 Diode I-V Curve

6. Connect diode: anode → FORCE HI, cathode → FORCE LO (Rear terminals).
7. Source Mode: Voltage Sweep, Start=0, Stop=1.0 V, Step=0.01 V.
8. Compliance=0.05 A, Source Delay=0.05 s, Input Jacks: Rear.
9. Confirm VISA address. Press RUN.

7.2 MOSFET Transfer Curve

10. Open FET window. Select Transfer Curve (Id vs Vg).
11. Gate: Start=0, Stop=8 V, Step=0.1 V, Compliance=0.01 A, Source Delay=0.10 s.
12. Drain: Op Mode = Voltage Bias, Bias=1.5 V, Compliance=0.5 A.
13. Press RUN. Status shows Acquiring… and the curve builds live.
14. On completion, data auto-saves. Enable Log Y Scale for subthreshold analysis.

✓ Tip: For a family of Transfer Curves at multiple Vds: set Drain Op Mode to Voltage Step, configure Drain Start/Stop/Step to the desired range. Each step produces one curve labelled Vds = X V.

7.3 MOSFET Output Curve

15. Select Output Curve (Id vs Vds).
16. Drain: Start=0, Stop=15 V, Step=0.2 V, Compliance=0.5 A.
17. Gate: Voltage Step, Start=2 V, Stop=6 V, Step=1 V. This produces 5 curves: Vgs = 2, 3, 4, 5, 6 V.
18. Press RUN.

7.4 Q-Point Analysis on Output Curves

19. Complete the Output Curve measurement (Section 7.3).
20. In the Q-Point panel, tick Enable.
21. Set Vdd=20 V, RL=100 Ω. Select the Vg=4 V curve from the dropdown.
22. Press Apply. Load line and Q-point appear on the chart.
23. Read Vds_Q, Id_Q, Pd_Q from the result bar and annotation box.
24. Press Save PNG to export the annotated chart.

7.5 Hysteresis Measurement

25. In the FET window, tick Dual Sweep in the Gate row (Transfer mode) or Drain row (Output mode).
26. Configure and press RUN. Both the forward and reverse passes are stored as one curve.
27. Enable the Hysteresis extractor in the Parameter Extraction panel and click Run Analysis.

ℹ Note: For TFTs and organic semiconductors, the Transfer Curve in Log Y Scale is particularly diagnostic for hysteresis from trap states.

7.6 Stability Cycling (IV Mode)

28. Configure a Voltage Sweep as usual.
29. In the Cycling & Stability card, tick Enable Cycling. Set Cycles=10, Rest=30 s.
30. Optionally add cycle names in the text box (one per line).
31. Press RUN. The sweep repeats 10 times with 30 s rest between cycles.
32. The chart shows all cycles in gradient colours with a Mean±σ band after cycle 2.
33. Export → Cycle CSV for wide-format data with one column per cycle.

7.7 Bias Stress Stability (FET Mode)

34. Set up a Transfer Curve measurement.
35. Expand Cycling & Bias Stress. Enable Cycling (Cycles=5), Enable Stress.
36. Set Terminal=Gate, Stress V=20 V, Duration=60 s, Measure every=10 s.
37. Press RUN. The sequence is: Sweep → Stress 60 s → Sweep → Stress 60 s → … × 5.
38. Drain current during stress is displayed in real time in the status label.
39. After completion, run Parameter Extraction to see ΔVth across cycles.

7.8 Live Probe Check with DMM

40. Open DMM mode from the Mode Selector.
41. Select Continuity. Touch probes to the device pads.
42. CLOSED (green) confirms a good ohmic contact. OPEN (red) indicates a problem.
43. Switch to DC Voltage or DC Current for quick bias-point verification.

8. Troubleshooting

Symptom	Resolution
VISA error / instrument not found	Open NI-MAX or Keysight Connection Expert, verify the resource string. Check USB cable. Reinstall NI-VISA if needed.
TSP-Link not online error	Check TSP-Link RJ-45 cable. Verify Node ID=1 on Master and Node ID=2 on Slave via MENU → System → TSP-Link Node.
All current values flat at compliance	Increase Compliance. Raise Source Delay to 0.2 s. Run Compliance pre-check to diagnose shorted device.
Noisy / non-monotonic Transfer Curve	Increase Gate Source Delay to 0.15–0.20 s. Enable Settling Check with Tol=1%. Increase NPLC to 5–10.
Blank plot after sweep	Click Auto Scale. Confirm Current (A) is ticked on Y1 in the Axis Selector.
Q-point shows No intersection	Increase Vdd or reduce RL so the load line crosses the saturation region.
Auto-save failed warning	Create the save folder manually, or click … in the header to select a writable directory.
Sweep timeout error	Reduce Points, reduce NPLC, or reduce Source Delay.
FET Export CSV empty after cycling	This was a bug in earlier builds. Update to the latest version of the script.
Parameter extraction returns N/A	Vth and SS require at least 4 data points with non-zero current. Ensure the sweep covers the subthreshold region.
DMM auto-disconnects repeatedly	Check VISA address is correct. Check USB cable. Ensure no other application holds the VISA session.
Ghost overlay not showing	Click Ghost: ON button in the IV toolbar. The ghost is only stored if data existed before the most recent RUN.

9. Quick Parameter Reference

9.1 IV Mode Defaults

Parameter	Default	Notes
Source Mode	Voltage Sweep	
Start / Stop / Step	0 / 5 V / 0.1 V	
Points	51	Auto-calculated from step
Compliance	0.01 A	10 mA — increase for higher-current devices
Source Delay	0.1 s	
NPLC	1	20 ms per point at 50 Hz
Auto Zero	Once	
Input Jacks	Rear	
Sense Mode	2-Wire	
Avg / point	1	No averaging by default
Cycling	Disabled	

9.2 FET Mode Defaults

Terminal	Operation Mode	Key Defaults
Gate	Voltage Linear Sweep (Transfer) / Voltage Bias (Output)	Start=0, Stop=8 V, Step=0.1 V, Comp=0.01 A, Delay=0.05 s
Drain	Voltage Bias (Transfer) / Voltage Linear Sweep (Output)	Bias=10 V, Comp=0.5 A, Delay=0.05 s
Source	GNDU	Not configurable when assigned to GNDU

Appendix A — Glossary

Term	Definition
Compliance	Maximum current (V source) or voltage (I source) the SMU will allow — acts as a protective clamp.
Cycling	Repeating the same sweep N times to measure device stability over time or stress cycles.
Bias Stress	Applying a sustained fixed voltage to a terminal (typically gate) to accelerate threshold voltage shift.
Ghost Overlay	A faded copy of the previous measurement displayed behind current data for visual comparison.
GNDU	Ground Unit — the chassis ground reference terminal of the Keithley 2450.
gm	Transconductance: dId/dVgs (Siemens). Indicates small-signal gain from the transistor.
Id	Drain current — the primary measured quantity in FET characterisation.
Ion/Ioff	Ratio of maximum to minimum drain current across the measured gate voltage range.
Mean±σ	Statistical band showing the mean and ±1 standard deviation across all completed cycles.
NPLC	Number of Power Line Cycles per measurement. 1 NPLC = 20 ms at 50 Hz.
Pd	Power dissipation at the Q-point: Pd = Vds × Id (watts).
Q-point	Quiescent operating point — steady-state DC bias conditions for a circuit.
SMU	Source Measure Unit — simultaneously sources voltage or current and measures the complementary quantity.
SS	Subthreshold Swing (mV/decade) — gate voltage needed to change drain current by one decade below Vth.
TSP	Test Script Processor — scripting language built into Keithley 2450 instruments.
TSP-Link	Keithley RJ-45 instrument bus allowing multiple SMUs to be synchronised from a single VISA connection.
VISA	Virtual Instrument Software Architecture — standard interface layer over USB, GPIB, LAN.
Vth	Threshold voltage — gate voltage above which the MOSFET begins to conduct in the linear region.

— End of Manual —

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